PHARMACOVIGILANCE DOCUMENT

ADR Identification Basics: Clinical Case Scenario Analysis & Causality Assessment

1. INTRODUCTION TO ADVERSE DRUG REACTION (ADR) IDENTIFICATION

1.1 Definition and Scope

An Adverse Drug Reaction (ADR), as defined by the World Health Organization (WHO), is "a response to a drug which is noxious and unintended, and which occurs at doses normally used in man for the prophylaxis, diagnosis, or therapy of disease, or for the modification of physiological function."

In clinical practice and post-marketing surveillance, distinguishing between an Adverse Event (AE) and an Adverse Drug Reaction (ADR) is fundamental:

- **Adverse Event (AE):** Any untoward medical occurrence in a patient or clinical investigation subject administered a pharmaceutical product, which does not necessarily have a causal relationship with this treatment. An AE can be any unfavorable and unintended sign, symptom, or disease temporally associated with the use of a medicinal product, whether or not considered related.
- **Adverse Drug Reaction (ADR):** An adverse event where a reasonable causal relationship between the medicinal product and the event is suspected or demonstrated.

1.2 Classification of Adverse Drug Reactions

Understanding ADR classification systems helps clinicians and safety evaluators predict, recognize, and manage drug toxicity:

Rawlins-Thompson Classification

- **Type A (Augmented):** Dose-dependent, predictable based on primary or secondary pharmacology. Examples: Hypoglycemia from insulin, bradycardia from beta-blockers, bleeding from anticoagulants. Management: Dose reduction or drug withdrawal.
- **Type B (Bizarre / Idiosyncratic):** Non-dose-dependent, unpredictable, driven by immunological hypersensitivity or genetic polymorphism. Examples: Anaphylaxis with penicillin, Stevens-Johnson Syndrome (SJS) with carbamazepine, drug-induced liver injury (DILI) from isoniazid. Management: Immediate discontinuation; future avoidance.
- **Type C (Chronic / Chemical):** Dose and time-dependent; associated with long-term therapy. Examples: Osteonecrosis of the jaw from bisphosphonates, adrenal suppression from chronic corticosteroids.
- **Type D (Delayed):** Time-dependent; becomes apparent long after drug exposure. Examples: Teratogenesis (phocomelia from thalidomide), secondary malignancies.
- **Type E (End-of-Use / Withdrawal):** Occurs shortly after withdrawal of a drug. Examples: Rebound hypertension following clonidine cessation, opioid withdrawal syndrome.
- **Type F (Failure of Therapy):** Unexpected lack of efficacy; often related to drug interactions, resistance, or quality defects.

2. METHODOLOGY OF CLINICAL CAUSALITY ASSESSMENT

Causality assessment is the systematic evaluation of the likelihood that a specific drug caused an observed adverse clinical event.

2.1 Key Parameters in Causality Evaluation

- **Temporal Relationship:** The timing between drug initiation/dose escalation and symptom onset. Event must follow drug administration with a plausible latency period.
- **Dechallenge:** The response after drug discontinuation or dose reduction. Positive dechallenge occurs when the symptom resolves or improves upon stopping the drug.
- **Rechallenge:** The response after re-introducing the suspected drug. Positive rechallenge occurs when the exact symptom recurs upon re-exposure.
- **Alternative Etiologies:** Evaluating non-drug causes, such as underlying disease progression, concomitant infections, or lifestyle factors.

2.2 Standardized Causality Assessment Tools

A. The Naranjo Causality Scale

The Naranjo algorithm utilizes a questionnaire with 10 questions scored -1, 0, +1, or +2:

Question	Yes	No	Do Not Know
1. Are there previous conclusive reports on this reaction?	+1	0	0
2. Did the adverse event appear after the suspected drug was administered?	+2	-1	0
3. Did the adverse reaction improve when the drug was discontinued or a specific antagonist was administered?	+1	0	0
4. Did the adverse reaction reappear when the drug was re-administered?	+2	-1	0
5. Are there alternative causes (other than the drug) that could on their own have caused the reaction?	-1	+2	0
6. Did the reaction reappear when a placebo was given?	-1	+1	0
7. Was the drug detected in the blood (or other fluids) in concentrations known to be toxic?	+1	0	0
8. Was the reaction more severe when the dose was increased, or less severe when the dose was decreased?	+1	0	0
9. Did the patient have a similar reaction to the same or similar drugs in any previous exposure?	+1	0	0
10. Was the adverse event confirmed by any objective evidence?	+1	0	0

Score Interpretation: Score ≥ 9: Definite ADR | Score 5 - 8: Probable ADR | Score 1 - 4: Possible ADR | Score ≤ 0: Doubtful ADR

B. WHO-UMC System

- **Certain:** Plausible temporal line; no alternative cause; positive dechallenge; positive rechallenge or lab confirmation.
- **Probable / Likely:** Plausible temporal line; unlikely related to disease/other drugs; reasonable dechallenge response.
- **Possible:** Reasonable temporal sequence; could also be explained by disease or other drugs; dechallenge unclear.

- **Unlikely:** Temporal relationship improbable; underlying disease or other drugs provide plausible explanations.

3. PRACTICAL CLINICAL CASE SCENARIOS & ADR EVALUATION

CASE SCENARIO 1: CARDIOVASCULAR & RESPIRATORY MEDICINE

Patient Demographics: 58-year-old male (Mr. A.R.)

Chief Complaint: Persistent, dry, non-productive cough lasting for 3 weeks, disturbing night sleep.

Medical History: Primary Essential Hypertension diagnosed 2 months ago. No history of asthma, COPD, or GERD.

Current Medication Record: Lisinopril 20 mg PO ONCE daily (Initiated 6 weeks ago); Amlodipine 5 mg PO ONCE daily; Atorvastatin 20 mg PO ONCE daily.

Symptom Timeline

Week 0: Lisinopril initiated | Week 3: Dry throat tickle and cough | Week 6: Cough persistent, worse at night | Week 6.5: Lisinopril discontinued, switched to Losartan | Week 8: Cough completely resolved.

Diagnostic Clinical Evaluation

1. **Symptom Review:** Persistent tickling cough. Lungs clear; chest X-ray normal.
2. **Suspected Drug:** Lisinopril (ACE Inhibitor).
3. **Mechanism:** ACE inhibition prevents breakdown of bradykinin and Substance P in pulmonary tree, causing irritation.
4. **Causality (Naranjo):** Score 7 (Probable ADR).
5. **Action Plan:** Type A Reaction. Discontinue ACE inhibitor; substitute with Angiotensin II Receptor Blocker (Losartan).

CASE SCENARIO 2: ENDOCRINOLOGY & METABOLIC DISORDERS

Patient Demographics: 62-year-old female (Mrs. S.K.)

Chief Complaint: Severe watery diarrhea (4-5 loose stools daily), abdominal cramping, flatulence, and metallic taste.

Medical History: Type 2 Diabetes Mellitus (newly diagnosed 3 weeks ago).

Current Medication Record: Metformin Immediate Release (IR) 850 mg PO twice daily; Glimepiride 1 mg PO once daily.

Symptom Timeline

Day 0: Started Metformin IR + Glimepiride | Day 4: Mild nausea and abdominal bloating | Day 10: Progresses to watery diarrhea | Day 21: Metformin dose reduced to 500 mg Extended-Release (XR) once daily | Day 28-35: GI symptoms resolved.

Diagnostic Clinical Evaluation

1. **Symptom Review:** Watery loose stools, nausea, cramps. Stool culture negative.
2. **Suspected Drug:** Metformin.
3. **Mechanism:** Local osmotic effect in intestine, altered gut microbiota, and increased intestinal glucose utilization.
4. **Causality (Naranjo):** Score 6 (Probable ADR).
5. **Action Plan:** Type A Reaction. Titrate Metformin slowly using Extended-Release (XR) formulation with heavy meals.

CASE SCENARIO 3: NEUROLOGY & DERMATOLOGY (CRITICAL ADR)

Patient Demographics: 24-year-old male (Mr. T.V.)

Chief Complaint: Widespread painful erythematous skin rash, oral mucosal ulceration, conjunctival congestion, fever.

Medical History: Focal Epilepsy (diagnosed 2 weeks ago). No prior drug allergies.

Current Medication Record: Carbamazepine 200 mg PO twice daily (Initiated 14 days ago).

Symptom Timeline

Day 0: Carbamazepine started | Day 10: Fever (38.1°C), sore throat | Day 12: Maculopapular rash on chest/face | Day 14: Blistering, epidermal detachment (<10% BSA), hospital admission | Day 14: Carbamazepine stopped | Day 28: Re-epithelialization complete.

Diagnostic Clinical Evaluation

1. **Symptom Review:** Epidermal necrosis, mucosal erosions. Diagnosed as Stevens-Johnson Syndrome (SJS).
2. **Suspected Drug:** Carbamazepine.
3. **Mechanism:** Type IV hypersensitivity reaction involving cytotoxic T-cells; associated with HLA-B*1502 allele.
4. **Causality (Naranjo):** Score 6 (Probable ADR) / WHO Category: Certain.
5. **Action Plan:** Type B Reaction. Absolute contraindication. Mandatory ICSR reporting.

CASE SCENARIO 4: ONCOLOGY & HEMATOLOGY

Patient Demographics: 52-year-old female (Mrs. M.P.)

Chief Complaint: High-grade fever (38.9°C), chills, severe fatigue, painful oral mucositis.

Medical History: Invasive Ductal Carcinoma of breast (Stage IIB). TAC Chemotherapy cycle 1 received 10 days ago.

Current Medication Record: TAC Chemotherapy (Docetaxel, Doxorubicin, Cyclophosphamide).

Symptom Timeline

Day 1: TAC Chemotherapy | Day 8: Progressive fatigue | Day 10: Onset of fever (38.9°C), ANC = 350 cells/mm³ (Severe Neutropenia) | Day 10: IV antibiotics & G-CSF initiated | Day 15: ANC recovered (>1500 cells/mm³).

Diagnostic Clinical Evaluation

1. **Symptom Review:** Febrile Neutropenia post-cytotoxic chemotherapy.
2. **Suspected Drug:** TAC Chemotherapy Regimen.
3. **Mechanism:** Direct cytotoxic destruction of rapidly dividing hematopoietic progenitor cells in bone marrow.
4. **Causality (Naranjo):** Score 7 (Probable ADR).
5. **Action Plan:** Type A Reaction. Prophylactic G-CSF (Filgrastim) in subsequent cycles.

4. ADR REPORTING & ICSR WORKFLOW

4.1 Individual Case Safety Report (ICSR) Mandate

The 4 Minimum Criteria for a Valid ICSR under ICH E2B guidelines:

- **1. Identifiable Reporter:** Physician, pharmacist, nurse, or patient.
- **2. Identifiable Patient:** Initials, patient ID, age, or gender.
- **3. Suspected Adverse Event:** MedDRA coded term.
- **4. Suspected Medicinal Product:** Brand or generic name.

4.2 Standardized Reporting Form Checklist

Includes Patient Details, Suspected Drug Information (dose, lot #, dates), Concomitant Medications, Adverse Event Details (onset, seriousness criteria), and Causality Assessment.

5. SUMMARY COMPARISON OF CASE SCENARIOS

Case	Primary Symptom	Suspected Drug	Type	Naranjo	WHO	Primary Action
1	Dry cough	Lisinopril	Type A	7 (Probable)	Probable	Stop ACEi; switch to Losartan
2	Watery diarrhea	Metformin	Type A	6 (Probable)	Probable	Switch to XR; titrate dose
3	Severe rash / SJS	Carbamazepine	Type B	6 (Probable)	Certain	Stop drug; ICU care; HLA test
4	Febrile Neutropenia	TAC Chemotherapy	Type A	7 (Probable)	Probable	Antibiotics + G-CSF prophylaxis